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**COURSE NAME : SOFTWARE  
ENGINEERING**

### **ASSESSMENT TOOL - 02**

**Title: AI-Based Smart Construction Project Monitoring and Risk Assessment Platform**

#### **1. Project Vision and Stakeholders**

##### **Project Vision**

To develop an AI-powered construction monitoring platform that enables real-time project tracking, risk assessment, safety monitoring, and predictive analytics using AI, IoT sensors, drones, and Building Information Modeling (BIM).

##### **Project Objectives**

- Monitor construction progress continuously.
- Detect safety violations using AI.
- Predict project delays.
- Optimize resource allocation.
- Track equipment utilization.
- Generate project performance dashboards.
- Assess construction risks.
- Improve overall project management efficiency.

##### **Key Stakeholders**

| <b>Stakeholder</b>     | <b>Role</b>                                     |
|------------------------|-------------------------------------------------|
| Project Manager        | Monitors project progress and schedules         |
| Site Engineers         | Update construction progress and resolve issues |
| Construction Workers   | Execute construction tasks safely               |
| Safety Officers        | Monitor safety compliance and hazards           |
| Clients                | Track project status and milestones             |
| AI/ML Development Team | Develop AI models for prediction and monitoring |
| IoT & Drone Providers  | Supply real-time site data                      |

| <b>Stakeholder</b>   | <b>Role</b>                                   |
|----------------------|-----------------------------------------------|
| System Administrator | Manage users, security, and maintenance       |
| Project Sponsor      | Provides funding and oversees project success |

## **2. Epics and User Stories**

### **Epic 1: User Authentication**

#### **User Story 1**

As a project manager,  
I want to log in securely,  
so that I can access project information safely.

### **Epic 2: Project Monitoring**

#### **User Story 2**

As a site engineer,  
I want to update construction progress,  
so that project status remains current.

#### **User Story 3**

As a project manager,  
I want to view a real-time project dashboard,  
so that I can monitor overall progress.

### **Epic 3: AI Risk Assessment**

#### **User Story 4**

As a project manager,  
I want AI to predict project delays,  
so that I can take preventive action.

#### **User Story 5**

As a safety officer,  
I want AI to detect safety violations,  
so that accidents can be prevented.

### **Epic 4: Resource Management**

#### **User Story 6**

As a site engineer,  
I want to monitor equipment utilization,  
so that resources are used efficiently.

## **Epic 5: Reporting**

### **User Story 7**

As a client,  
I want to receive project performance reports,  
so that I can monitor project completion.

### **User Story 8**

As a project manager,  
I want to view project risk reports,  
so that I can make informed decisions.

## **3. Acceptance Criteria (Given–When–Then)**

### **User Story 1 – Login**

Given the project manager has valid credentials  
When the login information is entered  
Then the system should display the project dashboard.

### **User Story 2 – Update Progress**

Given the engineer is logged in  
When project progress is submitted  
Then the system should update the project status successfully.

### **User Story 3 – Dashboard**

Given project data exists  
When the dashboard is opened  
Then current project progress and KPIs should be displayed.

### **User Story 4 – Delay Prediction**

Given project schedule data is available  
When AI analyzes the data  
Then predicted delays and recommendations should be shown.

### **User Story 5 – Safety Detection**

Given AI receives images from cameras or drones  
When a safety violation is detected  
Then an alert should be generated immediately.

### **User Story 6 – Equipment Monitoring**

Given equipment tracking data is available

When utilization is analyzed

Then equipment usage statistics should be displayed.

### **User Story 7 – Project Report**

Given project information exists

When the client requests a report

Then the latest project report should be generated.

### **User Story 8 – Risk Report**

Given AI has completed risk analysis

When the manager opens the risk module

Then project risks should be displayed with severity levels.

## **4. Product Backlog (Prioritized)**

| <b>Priority User Story</b> |                              | <b>Business Value</b> |
|----------------------------|------------------------------|-----------------------|
| High                       | Secure Login                 | Very High             |
| High                       | Construction Progress Update | Very High             |
| High                       | Project Dashboard            | Very High             |
| High                       | AI Delay Prediction          | Very High             |
| Medium                     | AI Safety Detection          | High                  |
| Medium                     | Equipment Monitoring         | High                  |
| Low                        | Project Performance Report   | Medium                |
| Low                        | Risk Assessment Report       | Medium                |

## **5. Sprint Backlog (Sprint 1)**

### **Sprint Duration**

**2 Weeks**

| <b>User Story</b> | <b>Development Tasks</b>                              |
|-------------------|-------------------------------------------------------|
| Secure Login      | Design login page, authentication, session management |

| User Story                   | Development Tasks                                     |
|------------------------------|-------------------------------------------------------|
| Construction Progress Update | Create progress entry form, database integration      |
| Project Dashboard            | Design dashboard UI, display project metrics          |
| AI Delay Prediction          | Develop prediction model, integrate schedule analysis |

### 6. Story Point Estimation (Fibonacci)

| User Story             | Story Points | Justification                                  |
|------------------------|--------------|------------------------------------------------|
| Secure Login           | 3            | Basic authentication functionality             |
| Progress Update        | 5            | Form validation and database operations        |
| Project Dashboard      | 5            | Data visualization and integration             |
| AI Delay Prediction    | 8            | AI model development and prediction logic      |
| AI Safety Detection    | 13           | Computer vision and real-time alert generation |
| Equipment Monitoring   | 8            | IoT integration and tracking                   |
| Project Report         | 5            | Report generation and export                   |
| Risk Assessment Report | 8            | Risk analysis and visualization                |

Estimation Technique: Fibonacci sequence (1, 2, 3, 5, 8, 13) based on feature complexity, uncertainty, and implementation effort.

### 7. Sprint Goal and Expected Deliverables

#### Sprint Goal

Develop the core construction monitoring features that allow project managers to securely log in, update construction progress, monitor project status through a dashboard, and receive AI-based project delay predictions.

#### Expected Deliverables

- ✔ Secure User Login Module
- ✔ Construction Progress Management
- ✔ Real-Time Project Dashboard
- ✔ AI-Based Delay Prediction Feature
- ✔ Database Integration
- ✔ Basic User Interface

-  Initial Testing and Bug Fixes

This first sprint delivers a functional prototype capable of monitoring construction progress and providing AI-assisted project insights for better decision-making.